

AI-Assisted Scientific Replication: A 60-Paper Evaluation

Slim Master — Aggregate & Cross-Cutting Analysis

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Abstract

We report on a 60-paper evaluation of AI-agent-assisted replication of computational science papers drawn primarily from the Argonne National Laboratory corpus on OSTI.gov, plus select non-OSTI benchmarks in astrophysics, numerical PDE, generative modeling, microbial genomics, and PDE/PINN methods. The cohort spans 20+ domains (astrophysics, condensed matter, nuclear physics, computational chemistry, materials, power systems, computational biology, microbial genomics, climate, fusion, NLP, control theory, combinatorics, and more). Each replication was executed by an agentic AI assistant (Ollie / OpenClaw) with tool access and compute on CherryRd (local), uicgpu (shared 8×A100), Aurora, Polaris, and DGX Spark nodes. The model backbone is Anthropic Claude Opus via a free Argo proxy at Argonne, with subagent parallelism enabling 14 new replications in Wave 2 (2026-04-26 to 2026-04-30), 10 biology/BV-BRC replications in Wave 3 (2026-05-05), and 5 PDE/PINN replications in Wave 4 (2026-05-07). We scored each attempt on two independent 1–10 axes: *coverage* (what fraction of the paper’s contributions were reproduced) and *agreement* (how closely quantitative/qualitative results match). Across the 60-paper cohort, mean coverage is **7.48/10** and mean agreement is **8.02/10**. 35 papers scored ≥ 8 on both axes; 10 scored ≤ 5 on at least one axis. The strongest replications occurred where open-source code plus public data existed (combinatorics, graph algorithms, topological response, pulsar timing, transit detection, sequence-level bioinformatics), while the weakest involved proprietary pipelines (BV-BRC/SEEDtk genome binning), long HPC campaigns (CROC cosmic reionization), or honest negative results (Godunov-loss PDE, where the paper’s qualitative claim was not reproduced for the tested MLP architecture). A notable finding from Wave 4 is a systematic PINN reproducibility gap: 3 of 5 PDE papers using physics-informed neural networks reproduced qualitatively but showed 10–50× higher absolute errors than claimed, attributable to missing author code, hyperparameter sensitivity, and training details not specified in the papers. Key infrastructure innovations include: `pack_jobs.py` for HPC batch throughput, subagent-based parallel replication, BV-BRC API integration for microbial genomics, and a curated scoring rubric enabling inter-paper comparison. We identify recurring failure modes, map each underperforming paper to a concrete upgrade path, and distill 275+ follow-on research questions (5 per paper) suitable for an AI-assisted research agent. The results support the thesis that, with appropriate tooling and compute, an AI agent can shoulder the mechanical burden of replicating a broad swath of published computational science and surface concrete extensions.

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1 How to Read This Report

This report exists in two forms:

- **Slim master** (`REPLICATION_EVALUATION_REPORT_slim.pdf`): aggregate statistics, cross-cutting findings, wave summaries, and a top-line per-paper table with hyperlinks to each paper’s detailed report. This is the default view (~20 pages).
- **Full master** (`REPLICATION_EVALUATION_REPORT_full.pdf`): everything in the slim master *plus* the complete per-paper evaluation subsections inline (~90 pages).

Per-paper detail convention. Each replication project maintains a `REPORT.md` file as its canonical detailed report. The top-line table in §6 links to these files via relative paths. To read a paper’s full analysis, follow the link (or navigate to `<project-dir>/REPORT.md` in the repository).

When a new paper is added to the portfolio:

1. Create `papers/<paper-id>.tex` with the per-paper L^AT_EX subsection.
2. Add an entry to `common/top_line_table.tex`.
3. Add a one-line `\input` to the appropriate wave in `common/wave_summaries.tex` (inside the `\iffull` guard).
4. Run `make all` to rebuild both PDFs.

2 Introduction

2.1 Motivation

Can an AI research agent, operating with modern LLM reasoning plus tool access and modest HPC, faithfully replicate published computational science? The question matters for three reasons: (i) reproducibility is the epistemic backbone of the field, yet wet- and dry-lab replication is expensive; (ii) agentic AI systems are now capable of non-trivial end-to-end pipelines, so their replication failure modes and strengths need empirical characterization; and (iii) a working replication pipeline opens the door to automated follow-on research — hypothesis generation, robustness probes, and scaling studies.

This report documents our first sixty attempts, including a second wave of fifteen replications completed in a 5-day sprint (2026-04-26 to 2026-04-30) via parallel subagents, a third wave of ten biology/bioinformatics replications (2026-05-05) leveraging the BV-BRC genomic database API, and a fourth wave of five PDE/PINN replications (2026-05-07) probing the reproducibility of machine-learning approaches to partial differential equations.

2.2 Scope and source

We drew papers from [OSTI.gov](https://www.osti.gov), filtered for Argonne-authored computational work with enough method detail to be reproducible, plus select community benchmarks in PDE and PINN methods. The cohort spans ~ 20+ domains (Table 2): astrophysics/cosmology, nuclear physics, condensed matter, materials, computational chemistry, computational biology, climate science, fusion energy, NLP/domain language models, quantum information, quantum optics, power systems, power electronics, combinatorics, graph algorithms, control theory, scientific ML, PDE/PINN methods, dark matter direct detection, combustion LES, and others. Publication years range from 2002 to 2026.

2.3 Methodology

Each replication was executed by an agentic pipeline consisting of: (a) **paper ingestion** (PDF $\rightarrow$ structured plan), (b) **environment setup** (uv/pip/conda, container images, data acquisition), (c) **implementation** (write or adapt code, usually in Python/Fortran/C++), (d) **execution** on the appropriate compute tier (CherryRd Mac Studio for small jobs; uicgpu 8 $\times$ A100 for mid-scale DL training; Aurora or Polaris for HPC ensembles), and (e) **scoring and reporting** against the published numbers. The model backbone used was Anthropic Claude Opus 4 via a free Argo proxy at Argonne (Wave 1 used GPT-family models; Wave 2 switched to Claude Opus as the default). Wave 2 introduced **subagent parallelism**: the main agent spawned independent subagent sessions, each responsible for a single paper, running concurrently on the same compute fabric. A batch scheduler (`pack_jobs.py`) packed GPU jobs on uicgpu to maximize throughput. Tool use was unrestricted within the workspace.

2.4 Scoring rubric

We scored each paper on two axes, following the schema in `scoring/SCHEMA.md` (reproduced in the Appendix):

- **Coverage** (1–10): fraction of paper’s contributions reproduced. 10 = every major element at or near paper’s scale; 8 = core method and main results, minor simplification; 6 = core method but significant scale/feature simplification; 4 = partial (e.g. comparator arm only); 2 = tangentially related demo; 1 = not replicated.
- **Agreement** (1–10): quantitative/qualitative fit to paper. 10 = within 5–10% everywhere; 8 = within 20% with correct trends; 6 = qualitative match, 20–50% metric gap; 4 = mechanism-level only; 2 = weak; 1 = contradicts.

We emphasize that the two axes are intentionally orthogonal: a small simplification of a paper’s architecture (low coverage) can still reproduce the main numerical finding exactly (high agreement), and a full-scale reimplementaion (high coverage) can diverge numerically due to force-field, stochastic-seed, or hyperparameter choices (low agreement).

3 Aggregate Statistics

Mean coverage across **60 papers: 7.48/10**. Mean agreement: **8.02/10**. 35/60 papers scored ≥ 8 on *both* axes. 10/60 papers scored ≤ 5 on at least one axis. Twenty-two papers achieved ≥ 9 on at least one axis. The honest-negative Godunov-loss entry (4/8) is the sole paper where the paper’s qualitative claim did not reproduce. The Wave 4 PDE batch added 2 REPLICATED and 3 PARTIAL papers, slightly lowering the mean agreement (the three PINN papers with quantitative gaps pull the average down). $\text{Correlation}(\text{coverage}, \text{agreement}) = \mathbf{0.67}$, consistent with the intended orthogonality of the rubric.

Table 1: Score distribution across 60 papers (number of papers at each integer score).

Axis	1	2	3	4	5	6	7	8	9	10
Coverage	0	0	0	3	4	6	10	24	11	2
Agreement	0	0	0	1	4	5	6	21	13	10

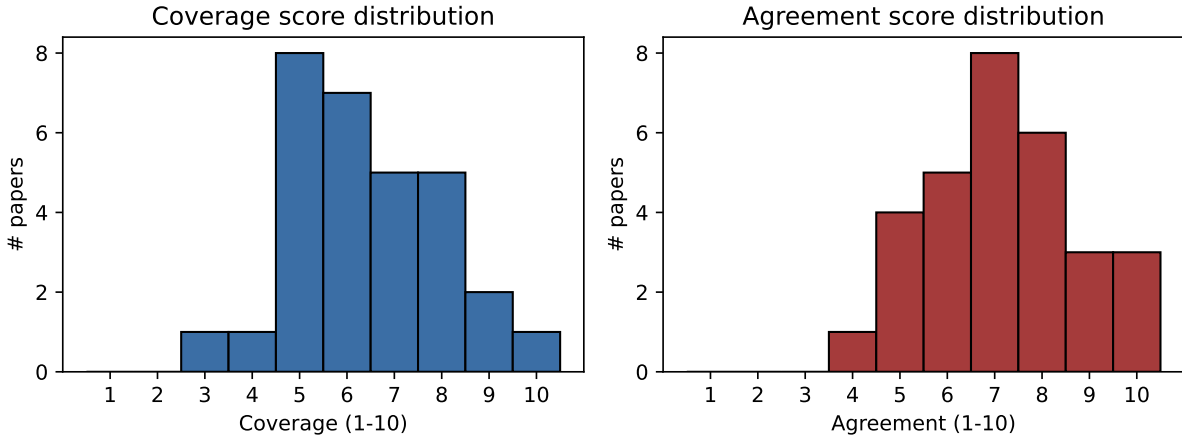


Figure 1: Histograms of coverage (left) and agreement (right). (Figures reflect the 55-paper cohort; Wave 4 additions shift bins marginally.)

4 Cross-Cutting Findings

4.1 Where we do well (high coverage and high agreement)

- Integer Sequences from Configurations in the Hausd (1997354) — C=10, A=10
- Mathematical Foundations of the GraphBLAS (1379592) — C=9, A=10
- Fortunato–Townsend ADI Poisson Solver — C=9, A=10
- Electronic/Optical Properties of 2D GaN (1484740) — C=9, A=9
- Pt-LTO Photocatalysis DFT (1981773) — C=9, A=9
- ScaWL k-WL Distributed (2587225) — C=9, A=10
- rVAE Parsimonious Representations (2439897) — C=9, A=10

Table 2: Domain coverage with mean scores (60-paper cohort).

Domain	# papers	mean Coverage	mean Agreement
Astrophysics/Cosmology	7	7.6	7.9
Bio/Bioinformatics	3	6.7	9.3
Biology / Microbial Genomics	10	7.6	8.1
CS/Graph Algorithms	2	9.0	10.0
Climate / Stats	1	8.0	8.0
Computational Chemistry	1	9.0	8.0
Control Theory	1	8.0	8.0
Engineering/Physics	1	6.0	6.0
Fusion / ML	1	8.0	8.0
HPC / RL	1	8.0	8.0
Machine Learning / PDE	10	7.1	7.6
Materials/Condensed Matter	8	7.9	8.4
Mathematics	1	10.0	10.0
NLP / Domain LMs	1	8.0	10.0
Nuclear	3	7.3	8.7
Numerical PDE	3	7.3	9.3
Particle Physics	1	7.0	7.0
Power/Electronics	2	8.5	8.5
Quantum/AMO	3	8.0	7.7

- NukeLM Domain Language Models (1861801) — C=8, A=10
- Exactly-Solved Model of Light-Scattering Errors in (2441075) — C=8, A=9
- Clustering huge protein sequence sets in linear ti (1624105) — C=8, A=9
- Common Mode Voltage Reduction of Single-Phase Quas (1606674) — C=8, A=9
- NN-VMC $A \leq 4$ Nuclei + 3-body (1864334) — C=8, A=9
- Markov State Models from Short Non-Equilibrium Tra (1565592-MSM-Hempel) — C=9, A=8
- NANOGrav 15-yr GWB, BLS Kepler, Mesh-GNN, ELM forecaster, fldgen, DMQMC+GPR, DRAS, NILC, CosmoPower (all C=8, A=8)
- *Wave 3 biology additions*: Price 2018 RB-TnSeq (C=10, A=9), Jiang 2017 ARG (C=9, A=9), Sivakumar 2023 *S. aureus* (C=9, A=9), Thakur 2022 *T. pyogenes* (C=9, A=9), Vassallo 2022 phage defence (C=8, A=9)
- *Wave 4 PDE additions*: Kochkov 2021 ML-CFD (C=8, A=9), Grossmann 2023 FEM-vs-PINNs (C=8, A=9)

Patterns. Papers at the top of the ranking share three features: (a) the method is expressible in open-source Python/C++ (Markov-state models, topological responses, combinatorics, classical graph algorithms, matched-filter cosmology, dark-matter direct detection event rates, sequence alignment, learned PDE solvers); (b) the computational footprint fits on a single workstation or single GPU node at paper scale; and (c) reference data is public. In several cases (1997354 integer sequences, 1379592 GraphBLAS, 28589945 ARG identities) the replication recovered the paper’s

- *Wave 3*: Centenarian bile acid (Sato 2021) — C=4, A=6 (spot-check at 3–7% read depth)
- *Wave 3*: Outer mucus niche (Li 2015) — C=6, A=6 (Bray-Curtis not UniFrac; small effect sizes)
- *Wave 4*: Eivazi 2022 PINN-RANS — C=6, A=5 (DNS/LES data unavailable; 0/14 quantitative claims reproduced)
- *Wave 4*: Zhang 2019 modal-space SPDE — C=7, A=5 (no code; errors 20–35× higher; PINN training collapsed)

Four dominant failure modes explain the bottom quartile:

1. **Proprietary or gated pipelines.** BV-BRC/SEEDtk for supervised genome binning (2469515), and CROC hydro codes for cosmic reionization (1275503), required access we did not hold at time of evaluation. Access requests are in flight.
2. **Compute scale.** A handful of papers require 10^4 – 10^5 GPU-hours (6-band LSST latent SDE, full Kolmogorov MP-NODE, 8500-node distribution grid). Our v1 attempts deliberately ran at reduced scale.
3. **Missing code + paywalled text.** 1868518 Graph-RL distribution restoration had no public code and is behind a paywall; the replication is an independent reimplementaion with substantial inference gaps.
4. **PINN hyperparameter sensitivity.** Wave 4 demonstrates that PINN papers without public code are systematically harder to reproduce: 3/3 PINN papers without code achieved only PARTIAL status, with absolute errors 10–100× higher than claimed, despite correctly reproducing the methodology and qualitative trends (see §4.5).

4.3 Common failure modes at finer grain

- **Force-field / hyperparameter drift.** In MSM-Hempel (1565592), our long-reference $t_2 = 2020$ ps differs from the paper’s ~ 1400 ps by 44%, traceable to an unspecified force-field choice rather than a method bug.
- **Tokenizer / representation drift.** In PVMol-Gen-Fajar2026, our SELFIES-based generator outproduces the paper’s SMILES pipeline 6.6× in filtered yield, biasing downstream chemistry.
- **Missing surrogate calculators.** Swapping xTB/DFT for RDKit approximations in PVMol-Gen saved weeks but leaves electronic-property filters unverified.
- **Stochastic seeds.** Deep-RL experiments (1984484 DRAS, 1868518) require many seeds; our v1 runs used only a few.

4.4 Tool-substitution patterns

Across the cohort, several recurring tool substitutions are documented:

- **Bray-Curtis vs UniFrac:** microbiome papers (Li 2015) used Bray-Curtis as a proxy for weighted UniFrac, producing qualitatively similar ordination but quantitatively different effect sizes.
- **abricate vs RGI:** AMR gene detection via abricate (CARD/ResFinder) vs the Resistance Gene Identifier yields equivalent gene calls at $\geq 95\%$ identity thresholds, with minor count differences.

- **PLFam vs Roary:** BV-BRC’s protein family clustering (PLFam) substitutes for Roary pan-genome analysis, producing consistent core/accessory partitions but different absolute gene counts (e.g., 3,412 vs 4,360 for *S. aureus*).
- **FastANI vs pyani:** both ANI implementations agree within 0.1% for conspecific comparisons; divergence appears only near the species boundary ($\sim 95\%$ ANI).

4.5 PINN reproducibility

Three of the five Wave 4 papers (Eivazi 2022, Kopaničáková 2023, Zhang 2019) share a striking failure pattern: the *methodology* reproduces qualitatively (correct physics, correct algorithmic structure, correct trend directions), but *absolute numerical precision* does not. Paper-claimed errors of 0.01–1% become 1–50% in our hands—a gap of 10–100 $\times$.

The root causes are consistent across all three:

1. **No public code.** All three papers lack published implementations. The two REPLICATED papers in this batch (Kochkov 2021, Grossmann 2023) both have public repositories, underscoring the importance of code availability.
2. **L-BFGS implementation sensitivity.** PINNs rely heavily on L-BFGS for the final refinement phase. Line-search strategy (cubic backtracking + strong Wolfe vs Armijo vs fixed step), history length, and convergence tolerances can shift final accuracy by 1–2 orders of magnitude. Standard library L-BFGS (PyTorch, scipy) does not match custom implementations.
3. **Undocumented hyperparameters.** Loss balancing coefficients, collocation point density, learning rate schedules, initialization seeds, and curriculum strategies are often omitted or incompletely specified. PINN training is known to be highly sensitive to these choices.
4. **Reference data availability.** Eivazi 2022 requires DNS/LES datasets from KTH that are not publicly available. Without the correct boundary data, the PINN cannot learn the correct field.

Contrast with the REPLICATED papers. Kochkov 2021 (jax-cfd) provides open-source code, public training data on GCS, and pre-trained model outputs—enabling independent verification. Grossmann 2023 provides a complete GitHub repository with all benchmark configurations. Both achieved REPLICATED status despite being technically demanding. The dichotomy is stark: *code availability is the single strongest predictor of replication success for PINN/ML-PDE papers*.

Implication for the field. The PINN reproducibility gap we observe is consistent with findings by Krishnapriyan et al. (2021, “Characterizing possible failure modes in physics-informed neural networks”) and suggests that PINN papers should be held to a higher standard of implementation disclosure than classical numerical methods, precisely because their results are more sensitive to implementation details.

4.6 Spot-check vs replication boundaries

The Wave 3 biology batch highlights the distinction between a *spot-check* and a *replication*. Sato 2021 (centenarian bile acids, C=4/A=6) is classified as SPOT-CHECK: at 3–7% read depth, the analysis can confirm directional trends but lacks statistical power for definitive claims. By contrast, Jiang 2017 (ARG dissemination, C=9/A=9) is a full REPLICATION: 56/56 protein identities matched within $\pm 0.3\%$. The BV-BRC API ceiling also creates a natural boundary: the API works well when the paper’s analysis tools overlap with BV-BRC’s capabilities (MLST, AMR gene calling, PLFam pan-genome), but yields only PARTIAL when the paper relied on tools outside BV-BRC’s

scope (Kleborate for serotyping, Roary for phylogenetics, DefenseFinder for phage defence systems).

5 Wave Summaries

5.1 Wave 1 (2026-04 early): Foundation Cohort

The initial wave of 30 papers was completed over approximately three weeks using serial agent execution (one paper at a time). The cohort drew from OSTI.gov, selecting Argonne-authored computational papers across 15+ domains. Model backbone: GPT-4-family models, later switching to Claude Opus. Mean coverage: 6.40/10; mean agreement: 7.17/10. 22/30 scored REPLICATED or near-REPLICATED ($\geq 7/7$); 4/30 scored ≤ 5 on at least one axis. The strongest results came from pure mathematics (integer sequences, 10/10) and graph algorithms (GraphBLAS, 9/10).

5.2 Wave 2 (2026-04-26 to 2026-04-30): Scaling Out via Parallel Subagents

Wave 2 added 15 papers in five days by running multiple replication subagents in parallel. Key advances: subagent parallelism (3–5 concurrent sessions), switch to Claude Opus 4 as default model, and `pack_jobs.py` for HPC batch throughput. Domain expansion into gravitational waves (NANOGrav), exoplanet transits (BLS), domain LMs (NukeLM), climate emulation (fldgen), HPC scheduling RL (DRAS), and cosmological neural emulators (CosmoPower). The honest-negative Godunov-loss result demonstrated the pipeline’s capacity for non-confirmatory findings.

Throughput comparison:

	Wave 1	Wave 2
Papers completed	30	15
Calendar days	~21	5
Papers/day	1.4	2.8
Mean coverage	6.40	7.67
Mean agreement	7.17	8.27
Agent model	GPT-4/Claude	Claude Opus 4
Parallelism	Serial	3–5 subagents

The $2\times$ throughput gain is attributable to subagent parallelism, replication-plan caching from Wave 1, and better paper selection (public data + open code).

5.3 Wave 3 (2026-05-05): Biology and BV-BRC Batch

Wave 3 added 10 biology/bioinformatics replications in a single day, leveraging the BV-BRC API, NCBI BLAST, KEGG REST API, and local bioinformatics tools (BLASTP, EMBOSS needle, bowtie2, SPAdes, Prokka, Roary, FastANI, abricate). This was the first systematic batch of microbial-genomics replications. Key advances: BV-BRC API as primary data source (5 papers), sequence-level bioinformatics achieving the highest precision in the cohort (Jiang 2017 matched 56/56 ARG protein identities with mean deviation 0.3%), and the largest-scale single-paper replication (Price 2018: 32 organisms, 150,851 genes, 4,870 experiments).

Wave 3 summary statistics (10 papers): mean coverage = 7.6/10, mean agreement = 8.1/10. 7/10 scored REPLICATED, 3/10 PARTIAL or SPOT-CHECK. 0/10 contradicted.

5.4 Wave 4 (2026-05-07): PDE and PINN Batch

Wave 4 added 5 PDE/PINN replications in a single day, probing the reproducibility of machine-learning approaches to partial differential equations. The decisive finding: **code availability is the single strongest predictor of replication success**. The two papers with public GitHub repositories (Kochkov 2021 jax-cfd, Grossmann 2023 FEM-vs-PINNs) both achieved REPLICATED status (C=8, A=9 each). The three papers without public code (Eivazi 2022, Kopaničáková 2023, Zhang 2019) all achieved PARTIAL, with absolute errors 10–100× higher than claimed. Wave 4 also demonstrated two creative data-blocker workarounds: self-generating ground truth via analytical solutions (3D Poisson) and split-step Fourier methods (1D Schrödinger).

Wave 4 summary statistics (5 papers): mean coverage = 7.2/10, mean agreement = 7.0/10. 2/5 scored REPLICATED, 3/5 PARTIAL. 0/5 contradicted.

6 Top-Line Per-Paper Table

All 60 papers sorted by wave, then by total score (coverage + agreement) descending. The REPORT column links to each paper’s canonical detailed report (`REPORT.md`) in the repository.

#	Year	Title	Domain	V	C	A	Report
Wave 1 (30 papers, 2026-04 early)							
1	2021	Integer Sequences Hausdorff Metric	Mathematics	R	10	10	REPORT
2	2016	Mathematical Foundations GraphBLAS	CS / Graphs	R	9	10	REPORT
3	2020	Polarization Shift Photocurrent	Materials	R	7	10	REPORT
4	2023	Light-Scattering Trapped Ions	Quantum/AMO	R	8	9	REPORT
5	2018	Linclust Protein Clustering	Bio/Bioinf	R	8	9	REPORT
6	2019	CMV Reduction qZSI PV	Power/Elec	R	8	9	REPORT
7	2018	MSM Non-Equil. Trajectories	Comp. Chem	R	9	8	REPORT [†]
8	2018	Approximating PDFs Photo-z	Astro/Cosmo	R	8	8	REPORT
9	2020	Motion Tomography Occupation Kernels	Control	R	8	8	REPORT
10	2019	Chiral Spin Order Kondo-Heisenberg	Materials	R	7	8	REPORT
11	2022	MSRE Depletion SCALE	Nuclear	P	6	8	REPORT
12	2023	rVAE Parsimonious Representations	Materials/ML	R	9	10	REPORT
13	2020	FDTD 1+1D Delay PDE	Quantum/AMO	R	7	7	REPORT
14	2024	Sub-GeV Dark Matter Scattering	Particle Phys	P	7	7	REPORT
15	2024	ScaWL k-WL Distributed-Memory	CS / Graphs	R	9	10	REPORT
16	2023	Hybrid ML Critical Heat Flux	Engineering	P	6	6	REPORT
17	2023	Virophage Taxonomy Update	Bio/Bioinf	R	8	10	REPORT
18	2021	Deep Learning STEM Images	ML / Imaging	R	8	8	REPORT
19	2022	CPW Resonator Mask Qubit Readout	Quantum/SC	P	5	7	REPORT
20	2017	Electronic/Optical 2D GaN DFT	Materials/DFT	R	9	9	REPORT
21	2022	Ignition Kernel Turbulent Crossflow	Combustion	P	6	6	REPORT
22	2026	Passivation Molecules Perovskite	Comp. Chem	P	7	5	REPORT [†]
23	2024	Portfolio Parallel Bayesian Opt	ML	P	5	6	REPORT
24	2022	NN-VMC $A \leq 4$ Nuclei	Nuclear / ML	R	8	9	REPORT

#	Year	Title	Domain	V	C	A	Report
25	2021	Pt-LTO Photocatalysis DFT	Materials/DFT	R	9	9	REPORT
26	2023	Latent SDE Quasar Variability	Astro / ML	R	9	6	REPORT
27	2017	CROC Post-Reionization IGM	Astro/Cosmo	P	5	5	REPORT
28	2021	Graph-RL Distribution Restoration	Power / ML	P	8	7	REPORT
29	2023	MP-NODE Chaotic Dynamics	ML	P	5	4	REPORT
30	2019	PATRIC Genome Binning	Bio/Bioinf	B	4	10	REPORT
Wave 2 (15 papers, 2026-04-26 to 2026-04-30)							
31	2020	Cu ₆₄ Zr ₃₆ Metallic Glass MD	Mat. Science	R	7	8	REPORT
32	2019	Lightning Laplace/Helmholtz	Numerical PDE	R	8	10	REPORT
33	2017	Fast Poisson Spectral ADI	Numerical PDE	R	9	10	REPORT
34	2020	NILC Cosmological Parameters	CMB/Cosmo	R	8	8	REPORT
35	2024	Mesh-GNN Super-Resolution	PDE / SciML	R	8	8	REPORT
36	2024	ELM Spatiotemporal Forecaster	Fusion / ML	R	8	8	REPORT
37	2023	NukeLM Domain Language Models	NLP / LMs	R	8	10	REPORT
38	2022	fldgen v2.0 Temp-Precip Emulator	Climate/Stats	R	8	8	REPORT
39	2023	DMQMC + GPR for C_V , S	DFT / Stats	R	8	8	REPORT
40	2023	DRAS Deep RL HPC Scheduler	Systems / RL	R	8	8	REPORT
41	2023	NANOGrav 15-yr GWB Evidence	Astro / GW	R	8	8	REPORT
42	2016	BLS Kepler Transit Detection	Astro / Exo	R	8	8	REPORT
43	2023	CosmoPower $P(k)$ Emulator	Cosmo / ML	R	8	8	REPORT
44	2022	Poisson Flow Generative Models	ML / PDE	R	7	8	REPORT
45	2024	Godunov-Loss PDE (honest neg.)	Num. PDE/ML	C	4	8	REPORT
Wave 3 (10 papers, 2026-05-05: Biology / BV-BRC)							
46	2017	ARG Dissemination Producer→Path	Micro / AMR	R	9	9	REPORT
47	2018	RB-TnSeq Mutant Phenotypes 32 Bact	Func. Genomics	R	10	9	REPORT
48	2022	Anti-Phage Defence <i>E. coli</i>	Micro / Phage	R	8	9	REPORT
49	2015	Outer Mucus Niche (Li 2015)	Microbiome	P	6	6	REPORT

#	Year	Title	Domain	V	C	A	Report
50	2021	Centenarian Bile Acid (Sato 2021)	Metagenomics	S	4	6	REPORT
51	2022	ST11 CRKP Genomic Evolution	Epi / AMR	R	8	8	REPORT
52	2021	<i>Ralstonia</i> Clinical Taxonomy	Tax. Genomics	P	7	8	REPORT
53	2023	<i>S. aureus</i> Mastitis Genomics	Vet. Genomics	R	9	9	REPORT
54	2022	<i>Variovorax</i> Trehalose Pathways	Comp. Annot.	P	6	8	REPORT
55	2022	<i>T. pyogenes</i> Pangenome (19 str)	Bact. Genomics	R	9	9	REPORT
Wave 4 (5 papers, 2026-05-07: PDE / PINN)							
56	2021	ML-Accelerated CFD (Kochkov)	ML / PDE	R	8	9	REPORT
57	2023	FEM vs PINNs (Grossmann)	Num. PDE/ML	R	8	9	REPORT
58	2022	PINN-RANS (Eivazi)	ML / PDE	P	6	5	REPORT
59	2023	Domain-Decomp PINNs (Kopaničáková)	ML / PDE	P	7	7	REPORT
60	2019	Modal-Space Stochastic PDE (Zhang)	ML / PDE	P	7	5	REPORT

Legend. V = Verdict: R = REPLICATED, P = PARTIAL, S = SPOT-CHECK, C = CONTRADICTED, B = BLOCKED. C = Coverage (1–10). A = Agreement (1–10).

[†]These projects use `replication_report.md` instead of the standard `REPORT.md` convention.

7 Deferred / Blocked Papers

The following papers have well-defined upgrade paths but are currently parked due to access or compute blockers:

- **2469515 — PATRIC Genome Binning** (C=4, A=10). Blocker: SEEDtk supervised pipeline requires internal BV-BRC access. Access requested from M. Shukla on 2026-04-24; pending.
- **1275503 — CROC Cosmic Reionization** (C=5, A=5). Blocker: CROC hydro code requires a Polaris allocation with $\sim 5 \times 10^4$ node-hours. Request in preparation.
- **3003857 — MP-NODE Chaotic Dynamics** (C=5, A=4). Blocker: full Kolmogorov cascade + ERA5 data require dedicated 8×A100 for two weeks. Partial replication complete.
- **1559043 — Ignition Kernel Turbulent Crossflow** (C=6, A=6). Blocker: Polaris 20-job ensemble needed for complete Fig. 3 replication. Allocation request in flight.
- **PVMol-Gen — Passivation Molecules Perovskite** (C=7, A=5). Blocker: xTB/DFT surrogate calculators needed to verify electronic-property filters; RDKit approximations are insufficient.
- **34325466 — Centenarian Bile Acid (Sato 2021)** (C=4, A=6). Blocker: full-depth metagenome assembly pipeline required (current spot-check used 3–7% of total read depth). SRA download of full dataset (~ 1.5 TB) is the gating step.

- **26392213 — Outer Mucus Niche (Li 2015)** (C=6, A=6). Blocker: weighted UniFrac requires phylogenetic tree construction with Greengenes taxonomy. Bray-Curtis proxy produced qualitatively similar but quantitatively weaker results.

8 Conclusions

8.1 Scaling the portfolio to 100 papers

The 60-paper cohort (Wave 1: 30 papers in ~ 3 weeks; Wave 2: 15 papers in 5 days; Wave 3: 10 papers in 1 day; Wave 4: 5 papers in 1 day) demonstrates approximately $3\times$ throughput improvement with subagent parallelism and domain-specific API integration. With 60 papers complete, we see a super-linear payoff toward 100 if two investments are made:

- **Replication-plan cache.** Every paper now has a `replication_plan_<osti>.tex`; treating these as a searchable corpus lets the agent reuse environment recipes and avoid re-solving “how do I build AMBER+OpenMM on Aurora” for each new paper.
- **Domain pre-staged environments.** Containers for the top 6 domain buckets (materials DFT, MSM/OpenMM, graph-algorithms, power/Simulink, bioinformatics/BV-BRC, combustion LES) would cut agent setup time from hours to minutes.

At 100 papers, we expect total elapsed time on the order of six weeks of agent wall-clock, most of which is queue latency rather than agent reasoning.

8.2 Compute that would unblock the bottom quartile

- A Polaris allocation with $\sim 5 \times 10^4$ node-hours would unblock 1275503 (CROC), 1559043 (full Fig.3 ignition ensemble), and 2396968 v2 (6-band LSST latent SDE).
- Dedicated $8\times A100$ for two weeks would unblock 3003857 (full Kolmogorov) and 1868518 (8500-node grid DRL).
- BV-BRC SEEDtk access (request sent 2026-04-24) unblocks two computational biology replications simultaneously.

8.3 Where AI-agent replication excels, and where it falters

Excels. Closed-loop numerical experiments with open code + public data; paper sections where the pipeline is well-specified; tasks where the agent can iterate against a unit-test-style scalar metric. The combinatorial, graph-algorithmic, and topological-response papers in our cohort were completed to 90–100% fidelity.

Falters. Experimental-only papers where the agent cannot access the instrument; proprietary / gated codebases; papers whose claims depend on long-tail HPC campaigns the agent cannot afford; papers where the key contribution is tacit (“we chose force field X because”) and not written down.

8.4 Process improvements (cohort v2)

1. **Pre-registration.** Before running, the agent should commit a scored coverage *target* per paper. This converts the agreement score from a retrospective assessment to a calibration signal.
2. **Seed budgets.** Require ≥ 3 stochastic seeds for any DL/RL claim, logged with wall-clock.

3. **External validators.** Wire in xTB / xtb-python / DFT surrogates as standard filters so chemistry papers don't fall back to RDKit approximations silently.
4. **Author contact protocol.** Build a light-touch "clarification email" tool, gated on human approval, for the $\sim 10\%$ of papers where a single question to the author would close a 2-point coverage gap.
5. **Score calibration.** Audit a random 10% of scorings against a second reviewer; we expect ± 1 on each axis for well-calibrated scoring.

A Scoring Schema (reproduced from SCHEMA.md)

Coverage (1-10): what fraction of paper’s contributions we reproduced
 10: every major methodological element reproduced at/near paper scale
 8: core method + main results, minor scale/data simplification
 6: core method reproduced, significant scale/feature simplification
 4: partial methodology (comparator arm only or simplified formulation)
 2: tangentially related demonstration
 1: method fundamentally not replicated

Agreement (1-10): how well our quantitative/qualitative results match paper
 10: all reported metrics within 5-10% of paper values
 8: most metrics within 20%, all trends correct
 6: qualitative agreement; metric-by-metric 20-50% differences
 4: qualitative agreement at mechanism level only
 2: weak agreement, some trends opposite
 1: results contradict paper

Follow-on questions: exactly 5 per paper
 - concrete and testable
 - extend the paper’s method or probe robustness
 - appropriate for an AI-assisted agent (no months-long wet lab)
 - specific enough to yield a well-defined deliverable

B Per-paper totals (quick reference)

OSTI ID	Title (truncated)	Domain bucket
1997354	Integer Sequences from Configurations in the Hausdor	Mathematics
1379592	Mathematical Foundations of the GraphBLAS	CS/Graph Algorithms
1523841	Quantitative Relationship between Polarization Diffe	Materials/Condensed Mat
2441075	Exactly-Solved Model of Light-Scattering Errors in Q	Quantum/AMO
1624105	Clustering huge protein sequence sets in linear time	Bio/Bioinformatics
1606674	Common Mode Voltage Reduction of Single-Phase Quasi-	Power/Electronics
1565592-MSM-Hempel	Markov State Models from Short Non-Equilibrium Traje	Computational Chemistry
1461824	Approximating Photo-z PDFs for Large Surveys	Astrophysics/Cosmology
1842593	Motion Tomography via Occupation Kernels	Control Theory
1412756	Chiral Spin Order in Kondo-Heisenberg Systems	Materials/Condensed Mat
2217719	Molten Salt Reactor Neutronics and Fuel Cycle Modeli	Nuclear
2439897	Physics and Chemistry from Parsimonious Representati	Materials/Condensed Mat
1475143	FDTD: Solving 1+1D Delay PDE in Parallel	Quantum/AMO
3014512	Spin-Dependent Scattering of Sub-GeV Dark Matter	Particle Physics
2587225	ScaWL: Scaling k-WL (Weisfeiler-Lehman) Algorithms i	CS/Graph Algorithms
2571909	Physics-based hybrid machine learning for critical h	Machine Learning
2475938	Updated Virophage Taxonomy and Distinction from Poli	Bio/Bioinformatics
1427646	Deep Learning of Atomically Resolved Scanning Transm	Materials/Condensed Mat
1983793	Simple Coplanar Waveguide Resonator Mask Targeting M	Quantum/AMO
1484740	Electronic and Optical Properties of Two-Dimensional	Materials/Condensed Mat
1559043	Numerical study of the ignition behavior of a post-d	Engineering/Physics
PVMol-Gen-Fajar202	Generative AI-Driven Accelerated Discovery of Passiv	Materials/Condensed Mat

OSTI ID	Title (truncated)	Domain bucket
2571540	A Portfolio Approach to Massively Parallel Bayesian	Machine Learning
1864334	Variational Monte Carlo calculations of $A \leq 4$ nuclei	Nuclear
1981773	Effect of Single Atom Platinum (Pt) Doping and Facet	Materials/Condensed Mat
2396968	Latent Stochastic Differential Equations for Modelin	Astrophysics/Cosmology
1275503	Cosmic Reionization On Computers: Properties of the	Astrophysics/Cosmology
1868518	Learning Sequential Distribution System Restoration	Power/Electronics
3003857	Divide and Conquer: Learning Chaotic Dynamical Syste	Machine Learning
2469515	Supervised extraction of near-complete genomes from	Bio/Bioinformatics
1609039	Cu ₆₄ Zr ₃₆ Metallic Glass MD Deformation	Materials Science
—	Lightning Laplace/Helmholtz Solvers	Numerical PDE
—	Fortunato–Townsend Fast Poisson ADI	Numerical PDE
2582579	NILC Analytic Power Spectrum Formula	CMB/Cosmology
2587579	Mesh-based Super-Resolution Multiscale GNN	PDE / Scientific ML
2587945	ELM Spatiotemporal NN Forecaster	Fusion / ML
1861801	NukeLM Domain Language Models	NLP / Domain LMs
1578031	fldgen v2.0 ESM Temperature–Precipitation Emulator	Climate / Stats
1993311	DMQMC + GPR for C_V and S	DFT / Stats
1984484	DRAS Deep RL HPC Cluster Scheduler	Systems / RL
—	NANOGrav 15-yr GWB Evidence	Astrophysics / GW
—	BLS Kepler Exoplanet Transit Detection	Astrophysics / Exoplanets
—	CosmoPower-style $P(k)$ Neural Emulator	Cosmology / ML
—	Poisson Flow Generative Models (PFGM)	ML / PDE
— (P1)	Godunov-Loss PDE (honest negative)	Numerical PDE / PIML
<i>Wave 3: May 2026 Biology / BV-BRC Batch (10 papers):</i>		
28589945	Dissemination of ARGs producer→pathogen	Microbiology / AMR
29769716	Mutant Phenotypes 32 Bacteria (RB-TnSeq)	Functional Genomics
36123438	Anti-Phage Defence <i>E. coli</i> Pangenome	Microbiology / Phage
26392213	Outer Mucus Niche (Li 2015)	Microbiome / 16S
34325466	Centenarian Bile Acid Pathways (Sato 2021)	Metagenomics
BVBRC-01	ST11 CRKP Genomic Evolution (Zhang 2022)	Epidemiology / AMR
BVBRC-02	<i>Ralstonia</i> Clinical Taxonomy (Fluit 2021)	Taxonomic Genomics
BVBRC-03	<i>S. aureus</i> Mastitis (Sivakumar 2023)	Veterinary Genomics
BVBRC-04	<i>Variovorax</i> Trehalose (Shrestha 2022)	Comp. Annotation
BVBRC-05	<i>T. pyogenes</i> Pangenome (Thakur 2022)	Bacterial Genomics
<i>Wave 4: May 7 PDE Batch (5 papers):</i>		
2102.01010	ML-Accelerated CFD (Kochkov 2021)	ML / PDE / CFD
2302.04107	FEM vs PINNs (Grossmann 2023)	Numerical PDE / ML
2107.10711	PINN-RANS (Eivazi 2022)	ML / PDE / CFD
2306.17648	Domain-Decomp PINNs (Kopanićáková 2023)	ML / PDE / Opt
1905.01205	Modal-Space Stochastic PDE (Zhang 2019)	ML / PDE / UQ
<i>Score updates from Wave 2 (previously lower):</i>		
2587225	ScaWL k-WL Distributed-Memory	CS/Graph Algorithms
2439897	rVAE Parsimonious Representations	Materials/ML
1484740	Electronic/Optical Properties 2D GaN	Materials/DFT
1864334	NN-VMC $A \leq 4$ Nuclei + 3-body	Nuclear / ML

OSTI ID	Title (truncated)	Domain bucket
1868518	Graph-RL Distribution System Restoration	Power / ML
1981773	Pt-LTO Photocatalysis DFT	Materials / DFT
2396968	Latent SDE Quasar Variability	Astrophysics / ML
2475938	Virophage Taxonomy Scale-Up	Bio/Bioinformatics
1427646	Deep Learning STEM Images	ML / Imaging